# Thien An Le, Ph.D.

**Catalysis and Process Development Engineer | Hydrogen Production | Ammonia Cracking | Catalyst Scale-up**

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## PROFESSIONAL SUMMARY

Chemical engineer (Ph.D.) with 10+ years of hands-on R&D in heterogeneous catalyst development for low-carbon energy: ammonia cracking to COx-free hydrogen, $CO/CO_2$ methanation and bio-oil upgrading. Track record of taking catalysts from formulation and bench screening through pilot-scale validation to **two industrial technology transfers** (Lotte Chemical Corporation; CES Co., Ltd.), backed by **10 granted patents** and 21 peer-reviewed publications. Combines catalyst synthesis, fixed-bed reactor testing, kinetic and stability analysis, and advanced characterization to diagnose performance and redesign materials. Experienced in delivering results to industrial partners and government sponsors through reports, proposals and IP. Seeking an engineering role in catalyst, materials or process development for hydrogen and energy applications.

## KEY ACHIEVEMENTS

- **Lab-to-plant delivery:** scaled ammonia-cracking catalysts from bench to pilot scale with an industrial partner, resulting in a technology transfer to **Lotte Chemical Corporation**; methanation catalyst work led to a second transfer to **CES Co., Ltd.**
- **Cost reduction:** developed Ru-free Ni and Co-Fe catalysts for ammonia cracking, including Ni/Al-Ce oxides that fully decompose NH3 near 550 °C.
- **Intellectual property:** 10 granted patents on hydrogen-production and methanation catalysts, including a patent family filed in the US, Japan, China and Korea.
- **Recognition:** ammonia-decomposition program selected among **Korea's National Top 100 R&D Achievements (2023)**; 21 publications (17 as first or co-first author) with 1,600+ citations.

## TECHNICAL SKILLS

- **Catalyst Development:** formulation and synthesis of supported noble and transition-metal catalysts (Ru, Ni, Co, Mo); sol-gel, co-precipitation, hydrolysis-precipitation, impregnation, hydrothermal and solvothermal routes; ammonolysis/nitridation; facet-controlled and doped oxide supports ($CeO_2$, $La_2O_3$-$CeO_2$, N-doped $CeO_2$); promoter design (Cs, Ce); control of dispersion, porosity and surface area via precursor, aging and calcination conditions
- **Reaction Engineering and Scale-up:** bench- and pilot-scale fixed-bed reactor operation with $NH_3$, $H_2$ and $CO/CO_2$ feeds at high temperature; online GC analysis; conversion and selectivity mapping; space-velocity (WHSV/GHSV) studies; reaction kinetics and apparent activation energy; turnover frequency (TOF) normalized to chemisorption site density; long-term stability and deactivation testing; bench-to-pilot scale-up with industrial partners
- **Characterization and Diagnostics:** XRD, XPS (CasaXPS deconvolution), TEM/STEM-EDS, SEM, BET/porosimetry, $H_2$ chemisorption, TPR/TPO/TPD/TPSR, in situ DRIFTS, FT-IR, Raman, UV-Vis, ICP-OES, TGA/DSC, synchrotron XAS (XANES/EXAFS); structure-performance and root-cause analysis
- **Modeling (collaborative):** DFT and microkinetic modeling with computational partners to identify rate-limiting steps and guide catalyst redesign
- **Technical Communication and Project Delivery:** technical reports, research proposals and patent applications for government and industry sponsors; technology transfer to industry; electronic lab records and data management; Microsoft Excel, Word and PowerPoint
- **Collaboration:** joint projects with industrial partners; international, multidisciplinary R&D teams; peer reviewer and journal editorial work

## PROFESSIONAL EXPERIENCE

**Senior Researcher | Korea Research Institute of Chemical Technology (KRICT), Daejeon, South Korea** **2019 - 2026**

*Hydrogen and C1 Gas Research Center | National hydrogen program: Integrated Original Technology for Use in Hydrogen Production and Storage*

- Led catalyst development for low-temperature ammonia cracking to COx-free hydrogen, designing Ru catalysts on engineered $CeO_2$-based supports ($CeO_2${100} facet-controlled, N-doped with Cs promotion, and $La_2O_3$-$CeO_2$ composite) to raise activity at lower temperature.

- Reduced precious-metal dependence by developing Ru-free Ni-based and Co-Fe catalysts (Appl. Catal. B 2022; Chem. Eng. J. 2024).
- Scaled catalyst preparation and reactor testing from bench to pilot scale with an industrial partner, leading to a **technology transfer to Lotte Chemical Corporation**.
- Planned and ran fixed-bed reactor test campaigns with online GC to quantify conversion, kinetics, apparent activation energy, space-velocity response and long-term stability; normalized intrinsic activity as TOF using $H_2$-chemisorption site densities for like-for-like catalyst comparison.
- Built structure-performance relationships from XRD, XPS (Ru electronic state), TEM, TPR/TPD/TPSR, in situ DRIFTS and synchrotron XAS, and worked with DFT partners to identify rate-limiting steps and redirect catalyst formulation.
- Delivered **9 granted patents** and 9 peer-reviewed publications; the ammonia-decomposition work was selected among **Korea's National Top 100 R&D Achievements (2023)**.
- Wrote project reports, research proposals and patent applications for government and industry sponsors, and coordinated deliverables with industrial partners.

**Ph.D. Researcher (Graduate Research Assistant) | Ajou University, South Korea** **2015 - 2019**

*C1 gas activation program: Novel Catalyst Technology for C1 Gas Activation*

- Developed nanostructured $CO/CO_2$ methanation catalysts by sol-gel, hydrolysis-precipitation and hydrothermal routes, tuning precursor chemistry, aging and calcination to control porosity, surface area and metal dispersion.
- Delivered catalysts active for low-temperature methanation; results supported a granted patent and a **technology transfer to CES Co., Ltd.**
- Investigated methanation mechanisms on tailored supports to guide catalyst design; produced 10 publications.
- Lecturer, Graduate School of International Studies, Ajou University (2017-2018).

**Graduate Research Assistant (M.S.) | Kyung Hee University, South Korea** **2012 - 2014**

- Ran biomass pyrolysis experiments to increase bio-oil yield and developed catalysts for bio-oil upgrading by hydrodeoxygenation; 2 publications.

## EDUCATION

**Ph.D., Energy Chemical Engineering**, Ajou University, South Korea — Aug 2019
**M.S., Chemical Engineering**, Kyung Hee University, South Korea — Feb 2014
**B.Eng., Chemical Engineering**, Industrial University of Ho Chi Minh City, Vietnam — Sep 2009

## PATENTS AND SELECTED PUBLICATIONS

**Patents:** 10 granted patents on catalysts for hydrogen production and methanation (9 from KRICT, 1 from Ajou University), including US Patent 12,251,684 B2 (2025) and patents in Japan, China and Korea.

**Publications:** 21 peer-reviewed articles (17 as first or co-first author); 1,620 citations; h-index 17 (Google Scholar, Oct 2026). Full list on Google Scholar.

1. **Le TA**, Kim YM, Han SJ, et al. Synergistic electronic modulation of Ru sites via N-doped $CeO_2$ and Cs promotion for high-efficiency $H_2$ generation from ammonia decomposition. *J. Mater. Chem. A* (2026).
2. **Le TA**, Kim YM, Han SJ, et al. Ru dispersed on $CeO_2${100} facets boosting the catalytic $NH_3$ decomposition for green $H_2$ generation. *Chem. Eng. J.* 493 (2024) 152503.
3. Do QC, Kim YM, **Le TA**, et al. Facile one-pot synthesis of Ni-based catalysts by cation-anion double hydrolysis method as highly active Ru-free catalysts for green $H_2$ production via $NH_3$ decomposition. *Appl. Catal. B* 307 (2022) 121167.
4. **Le TA**, Kim YM, Kim HW, et al. Ru-supported lanthania-ceria composite as an efficient catalyst for COx-free $H_2$ production from ammonia decomposition. *Appl. Catal. B* 285 (2021) 119831.

## AWARDS AND PROFESSIONAL SERVICE

- Korea National Top 100 R&D Achievements (2023); Chae Yung Bog Best Paper Award, KRICT (2021)
- Editorial Board Member and Guest Editor, *Catalysts* (MDPI) (2021-); peer reviewer for 20+ international journals; Nomination Reviewer, VinFuture Prize (2021-)